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Res2Next with attention mechanisms for malware classification based on feature visualization

Liangwei Yao ^{a b} , Hongliang Zhu ^a , Yang Xin ^a  [Show more](#) ▾ Cite  Add to Mendeley  Share | 10.1016/j.jisa.2025.104271 ↗[Access through your organ...](#)[Purchase PDF](#)**Article preview**

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Highlights

- Proposed a novel lightweight ACA attention mechanism to capture crucial information more intelligently and precisely.
- Designed various ensemble models by integrating different plug-and-play attention mechanisms to effectively enhance the performance of malware classification.
- Employed three visualization methods to provide strong interpretability support for malware classification.
- Conducted extensive experimental evaluations to effectively demonstrate the superior performance of the proposed ACA mechanism and the ensemble models.

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Abstract

Malware poses a significant threat to cybersecurity due to its diverse types, complex behaviors, and strong destructiveness. Accurately classifying malware is crucial for taking effective defense measures. However, traditional malware classification methods based on static and dynamic features face challenges such as poor adaptability, manual intervention, and low classification accuracy. Although improvements have been made with visualization-based image classification methods, they remain susceptible to interference information in deep feature extraction. To this end, this paper proposes an innovative malware classification framework that utilizes the feature visualization method to convert malware into RGB images, effectively preserving its rich features and avoiding reverse engineering. Afterward, a lightweight adaptive channel attention (ACA) mechanism is proposed, and ensemble models based on Res2NeXt that integrate various attention mechanisms are designed for deep feature extraction and classification. In addition, through t-SNE visualization, confusion matrix, and Grad-CAM heatmap display, the proposed Res2NeXt with ACA model as a typical example shows superior

performance in feature space distribution, classification accuracy, and focusing on crucial features. In summary, a series of experiments conducted on public datasets, MMCC and MaleVis, demonstrate that the attention mechanisms in the ensemble models can effectively guide the model to focus on crucial features, filter out interference information, and enhance classification effectiveness. Specifically, the ACA attention mechanisms significantly improve classification accuracy with minimal impact on the model's efficiency. The proposed framework achieves classification accuracy of up to 99.26% and 98.04%, respectively, surpassing the current state-of-the-art methods.

Keywords

Malware classification; Res2NeXt; Attention mechanism; Feature visualization; Lightweight ACA

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Introduction

As cybersecurity threats intensify, malware, one of the most common and destructive forms of cyber attacks, is growing at an unprecedented speed and complexity. According to the 2024 Mid-Year Cyber Threat Report released by SonicWall, the volume of malware has surged by 30% compared to the same period last year, with a massive 92% increase in May alone [1]. This significant growth highlights the severe threat that malware poses to cybersecurity. There is a wide variety of malware, mainly including viruses, worms, Trojans, spyware, and ransomware, which can be classified into numerous subcategories, each with different purposes such as stealing sensitive information, disrupting normal system operations, monitoring user behavior, or extorting ransom from victims. Because malware developers often utilize techniques like packing, encryption, polymorphism, and obfuscation to create different types or families of malware to evade existing security measures, the task of efficiently and accurately classifying large volumes of malware to devise effective countermeasures has become extremely complex.

Currently, the existing traditional malware classification methods mainly rely on extracting the file structure [2], [3], string sequence [4], N-gram [5], opcode [6], [7], audit log [8], control flow graph (CFG) [9], [10], function call graph (FCG) [11], [12], [13], [14], heterogeneous graphs [15], [16], [17], application programming interface (API) calls [18], [19], [20], network traffic [21], [22], [23], and combinations of the above features [24], [25], [26] to complete the malware classification task. Although these features have achieved significant improvement in malware classification, there are still some issues: for example, attackers can modify file structures or strings to evade detection (e.g., [2], [3], [4]); disassembling malware into bytecode is not only time-consuming but may also lead to information loss (e.g., [5]); decompilation is required, and the resultant code may be incomplete or contain errors (e.g., [6], [7]); generating graph structures can be complex or incomplete and requiring considerable computational resources (e.g., [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]); executing in specific virtual environments with high spatiotemporal overhead (e.g., [8], [13], [18], [19], [20], [21], [22], [23], [24], [25]); and the processes of

decompression, disassembling, and decompilation are cumbersome and inefficient (e.g., [26]).

To address the above issues faced by traditional malware classification methods and effectively counter the ever-evolving malware threats, researchers are exploring more advanced technological solutions. In recent years, classic convolutional neural networks (CNNs), such as VGG [27], ResNet [28], Inception [29], SqueezeNet [30], DenseNet [31], Xception [32], ResNext [33], MobileNet [34], EfficientNet [35], Res2Net [36], RegNet [37], HS-ResNet [38], and CoAtNet [39] have inspired the popularity of malware visualization techniques due to their outstanding performance in image classification tasks. Malware visualization methods typically read relevant information about the malware in binary or byte stream form, then convert the extracted features into grayscale or color images, and finally classify the images based on the classic CNNs or their improved versions, thereby achieving malware classification [40], [41], [42], [43], [44], [45], [46], [47], [48], [49]. However, traditional CNNs face issues in feature extraction, such as difficulty in focusing on crucial feature information, loss of a significant amount of valuable information, and insufficient ability to capture long-distance dependencies, which leads to the need for improved performance in malware classification.

How to more effectively visualize malware features, accurately capture crucial information in malware images, and better capture long-range dependency information to improve the performance of malware classification is an important and urgent issue that this paper aims to address. Therefore, this paper presents an innovative framework. The framework employs a feature visualization method that visualizes malware as three-channel RGB images, introduces a novel ACA mechanism, explores various attention mechanisms, and designs different ensemble models based on Res2Next that integrate multiple plug-and-play attention mechanisms for deep feature extraction and malware classification. By organically combining the visualized RGB image representation, attention mechanisms, and the Res2Next model, we significantly enhance the visualization capabilities of malware features, strengthen the sustained focus on crucial features, and expand the receptive fields of neurons, which makes it easier to capture long-distance dependency information. Experimental evaluation demonstrates that our framework strikes a good balance between effectiveness and efficiency in malware classification, reducing the impact of irrelevant or redundant information and effectively improving the accuracy of malware classification. In addition, this paper employs different visualization methods, including t-SNE [50], confusion matrices, and Grad-CAM [51], which help security analysts better understand the decision-making processes of the model and enhance its interpretability. In summary, the main contributions of this paper include:

- A novel lightweight ACA is proposed, which primarily focuses on calculating the channel relationships in feature maps to dynamically adjust weight distribution, thereby capturing crucial information more intelligently and precisely.
- Various ensemble models are designed by integrating different plug-and-play attention mechanisms based on the Res2Next model, fully leveraging the multi-scale representation capability of Res2Next and the crucial feature-capturing ability of attention mechanisms, effectively enhancing the performance of malware classification.

- A variety of visualization methods, such as t-SNE, confusion matrices, and Grad-CAM, are employed to intuitively showcase the two-dimensional (2D) space distribution of malware features, prediction results for each class, and the model's focus on specific regions of the images, providing strong interpretability support for malware classification.
- Extensive experimental evaluations are conducted on the MMCC and Malevis datasets, effectively demonstrating the superior performance of the proposed ACA mechanism and the ensemble models based on Res2Next that integrate various attention mechanisms.

The rest of this paper is organized as follows:

Section 2 reviews the related work on malware classification. Section 3 details utilized malware visualization method, proposed ACA mechanism, and designed ensemble models. Section 4 describes experimental designs and discusses the results. Finally, Section 5 presents the conclusion of the paper.

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This section provides an overview of commonly used feature analysis methods in malware classification. It categorizes these methods into two main types based on their manifestations: traditional non-visual feature analysis methods and visual feature representation methods. ...

Methodology

In this section, we elaborate on the complete process of malware classification. First, a feature visualization method based on color images was introduced for feature extraction and representation of malware. Next, an innovative lightweight ACA mechanism was proposed, aiming to efficiently capture complex channel dependencies, thereby achieving a precise focus on crucial features. Finally, ensemble models (i.e., Res2NeXt with ACA, Res2NeXt with SE [57], Res2NeXt with CBAM [58], Res2NeXt with ...

Experiments

In this section, the paper first provides a detailed introduction to the experimental environment, then describes the relevant evaluation metrics used to assess model performance, followed by an introduction to the two datasets used in the experiments and their preprocessing procedures, and finally compares and analyzes the experimental results. ...

Conclusion

This paper primarily focuses on research aimed at improving malware classification performance, proposing a classification framework that combines Res2NeXt with attention mechanisms based on visual RGB images. First, we employed a feature visualization method based on color images to extract and represent malware features. This method not only successfully preserves the rich features of malware but also eliminates tedious reverse engineering. Next, we provided a detailed introduction to the ...

CRedit authorship contribution statement

Liangwei Yao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Hongliang Zhu: Writing – review & editing, Visualization, Validation, Software, Resources, Investigation, Formal analysis, Data curation. **Yang Xin:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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References (71)

Weijie Han *et al.*

[MalInsight: A systematic profiling based malware detection framework](#)

J Netw Comput Appl (2019)

Yonghe Tang *et al.*

[Bhmdc: A byte and hex n-gram based malware detection and classification method](#)

Comput Secur (2023)

Yewei Zhen *et al.*

[A novel malware detection method based on audit logs and graph neural network](#)

Eng Appl Artif Intell (2025)

Minh Hai Nguyen *et al.*

[Auto-detection of sophisticated malware using lazy-binding control flow graph and deep learning](#)

Comput Secur (2018)

Zhen Liu *et al.*

[SeGDroid: An android malware detection method based on sensitive function call graph learning](#)

Expert Syst Appl (2024)

Diego Soi *et al.*

[Enhancing android malware detection explainability through function call graph APIs](#)

J Inf Secur Appl (2024)

Zhendong Wang *et al.*

[Fagnet: Family-aware-based android malware analysis using graph neural network](#)

Knowl-Based Syst (2024)

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